



AI Training Session: From Zero to Hero in 45 Minutes

A fast-paced introduction to AI fundamentals and practical applications.

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AI and you



Quick Poll

Who's using ChatGPT? Other AI tools?



Icebreaker

Wildest thing you've done with AI recently?



Don't be shy



The "iPhone Moment" of AI



November 2022

ChatGPT launches, 100M users in 2 months



Different Wave

Why this feels different from previous tech

"talking to the machine in your language"



Live Demo

Random task → ChatGPT does it in real-time

Our conversation today

Understanding the basics

What's actually happening when AI works

Foundation Building

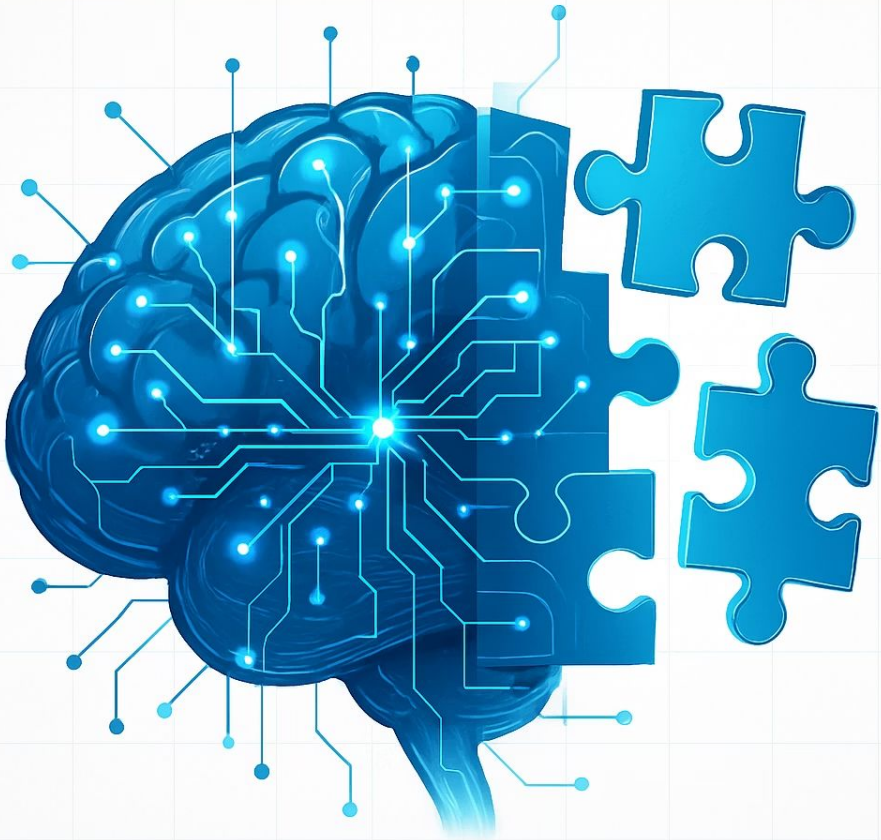
Not a tutorial, but fundamental concepts

Smarter Usage

Apply knowledge to any AI tool

Know what you're talking about





6 Essential Concepts

Decode AI Interactions

Understanding what happens when you chat with AI

Interactive Learning

Hands-on demonstrations of key concepts

Practical Insights

Knowledge you can apply immediately

Core Concept #1: Predicting the Next Word



Input Text

"The capital of France is..."



Pattern Recognition Engine

AI processes patterns in data

Trained on billions of words (or images, or sounds)



Prediction

AI completes with "Paris"

Most likely word vs training data

AI SYSTEM



The capital of
France is ...

P

A

F

Paris

WORD PREDICTION

Core Concept #2: Tokens (and context window)



More on tokens / context window

Context Window Evolution

Model	Max Tokens	Approx. Pages*	Notes
GPT-3.5	4,096	~13 pages	Default in ChatGPT free tier until early 2024
GPT-4 (2023)	8,192	~26 pages	ChatGPT Plus (early version), limited context
GPT-4 (128K)	128,000	~400 pages	GPT-4-Turbo, October 2023 release
GPT-4.1	~1M (est.)	~3,000+ pages	2025 estimate; full 1 million token context supported via OpenAI API (see below)



Real-Life Reference: Tokens to Pages

Tokens	Words (est.)	Pages (est.)	Real-Life Analogy
10,000	~7,500	~25 pages	Long-form article or white paper
128,000	~96,000	~350–400 pages	A full book (e.g., <i>Sapiens</i>)
256,000	~192,000	~750–800 pages	Two-volume textbook set
1,000,000	~750,000	~3,000+ pages	Entire Wikipedia + War & Peace

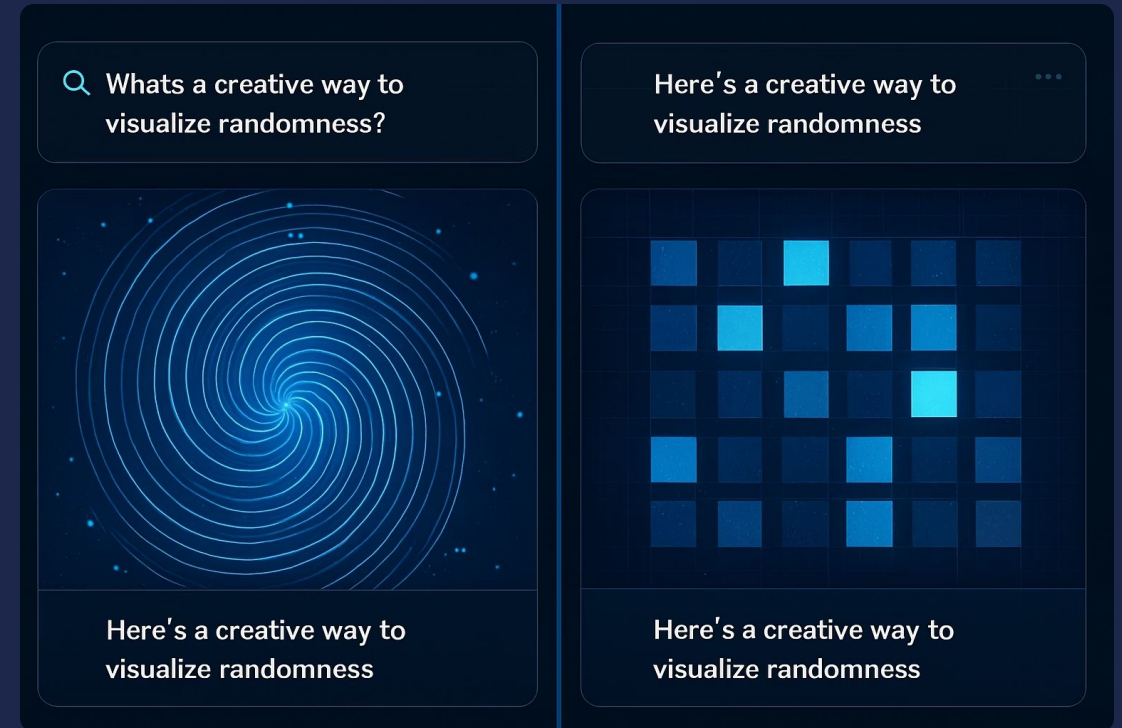
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Core Concept #3: Stochasticity

The Creativity Dial

Same input $\neq$ same output

- Enables creative responses
- Creates inconsistency
- Can be adjusted



Live demo: Ask same question twice, get different answers

Core Concept #4: Hallucinations

?

Fake legal cases

Context:

In a federal court in New York, attorney **Steven Schwartz** (of the law firm Levidow, Levidow & Oberman) used **ChatGPT** to help write a legal brief in a personal injury lawsuit against Avianca Airlines.

The Problem:

The brief cited **six court cases** that... simply **did not exist**. They were:

- *Varghese v. China Southern Airlines*
- *Martinez v. Delta Airlines*
- *Petersen v. Iran Air*
- ...and others, all sounding plausible but entirely fabricated.

ChatGPT even **doubled down** when asked for sources, providing **fake docket numbers** and judicial decisions. The model did what it does best: confidently synthesize realistic-sounding outputs without verifying any facts.



Confident answers

AI provides incorrect information but wants to make you happy



What to do ?

RAG + sources citation + dont believe in Santa Claus

Core Concept #5: Prompting is Everything

That's how you interact with the system. Like a JQL query - but it's your language, so it's easy to forget.

Like a very smart intern - who read all the books but has no knowledge of your implicit context.

Bad Prompt

"Write something"

Good Prompt

"Write professional email declining meeting, friendly tone, suggest alternatives"

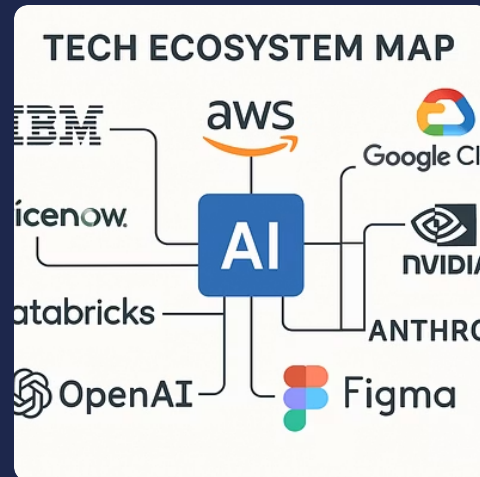
"I need to make an important decision. Here is the situation: (describe the situation). Here are the solutions I have identified so far: (list the solutions you've identified). I want you to act as an expert in identifying risk and help me see the second-order consequences of these solutions."

Core Concept #6: Multimodality

Multimodal AI: Inputs vs. Outputs

Input Modality	Possible Outputs
Text	Text, Code, Image, Audio, Video
Image	Text (description), Image (edit), Video (animation)
Audio	Text (transcription), Audio (voice clone), Video
Video	Text (summary), Audio (extraction), Video (edit)
Code	Code (completion), Text (explanation), Image (UI)
Mixed (e.g. PDF)	Text, Image, Code, Summary, Action Steps

Part 2: The AI Landscape



Who does what in the world of AI?

Foundation models and the rest

Foundation Models

- OpenAI (GPT-4 / o-3)
- Google (Gemini)
- Anthropic (Claude)
- Meta (Llama)
- xAI (grok)
- Chinese models



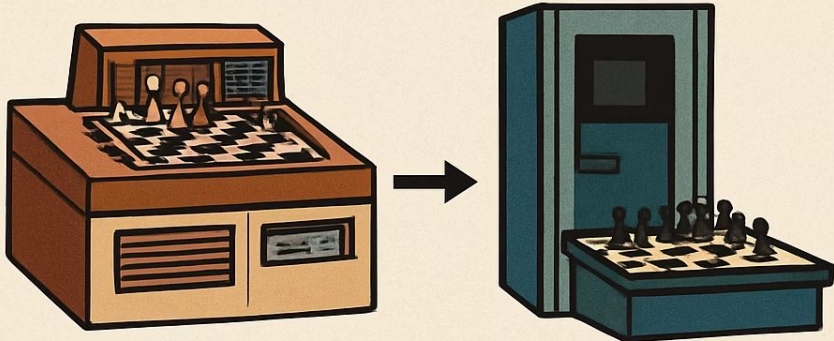
Interfaces & Wrappers

- ChatGPT $\neq$ GPT-4
- Perplexity
- Microsoft Copilot, Rovo

The Difference

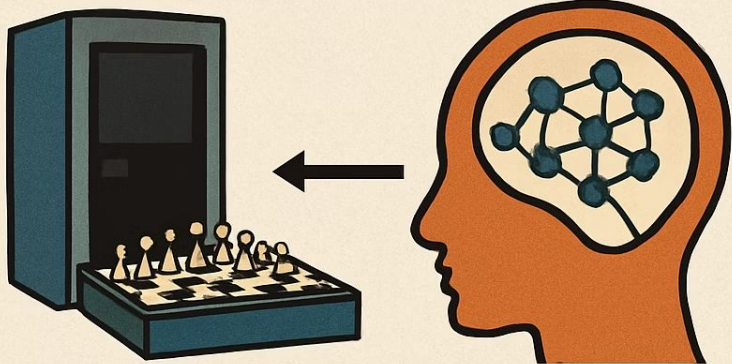
- Model = engine
- Chatbot = car you drive
- Agent = self-driving car

EVOLUTION TIMELINE



EARLY CHESS
COMPUTER

DEEP
BLUE



NEURAL
NETWORK

GENERATIVE
AI

Quick Evolution Story

1

Traditional AI (1950s-now)

Chess computers, spam filters, recommendations

2

Famous Milestones

DeepBlue (1997), AlphaGo (2016), GPT-3 (2020)

3

Generative AI (2020s-now)

Creates, converses, reasons



Part 3: What This Means for You

3

Key Areas

Superpowers, gotchas, applications

4

Golden Rules

For effective AI use

10

Minutes

Of critical thinking about AI impact

Superpowers You Now Have



Content Creation

Essays, emails, code, images, presentations



Research Assistant

Summarizing, explaining, brainstorming



Personal Tutor

Learning skills, personalized explanations



Gotchas You Need to Know



Confident Lies

Always verify important information (sources, deep search)



Biases

Trained on internet data with flaws (or master prompt)



Not Updated

Knowledge cutoffs, need to browse for recent events

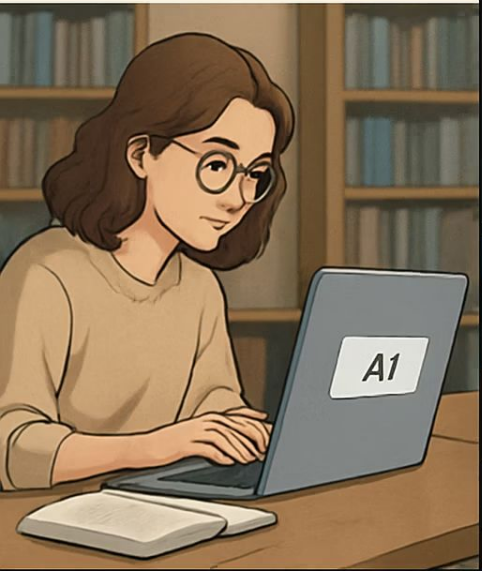


Privacy Concerns

Don't share (too) sensitive data (and then get off Insta)



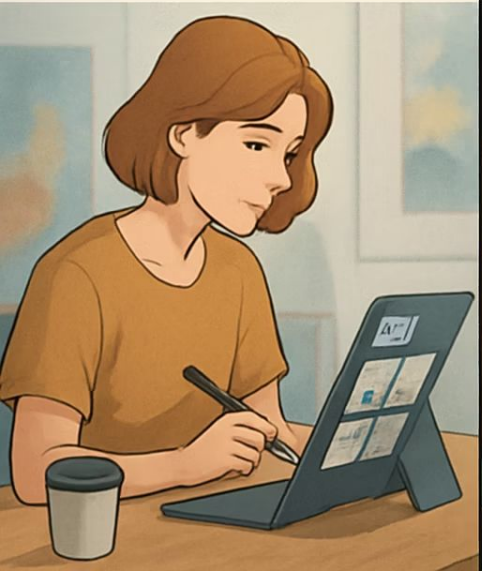
**STUDENT
RESEARCHING**



**BUSINESS PERSON
DRAFTING EMAIL**



**ARTIST
GENERATING IDEAS**



**EVERYDAY PERSON
MANAGING TASKS**



Real-World Applications by Role

Students

Research, writing assistance,
learning complex topics

Professionals

Email drafting, data analysis,
brainstorming, code review

Creatives

Ideation, content creation,
overcoming creative blocks

Everyone

Personal assistant for daily
tasks

The Golden Rules



1. Co-pilot, not autopilot

You're still in the driver seat



2. Trust but verify

Especially for facts and decisions



3. Iterate your prompts

First answer rarely best answer. Prompting = conversation + context



4. Understand the tool

Different models, different strengths

Major AI Debates and Ethical Considerations

Controversies

- Scaling vs. new paradigms
- Is AGI really coming? tomorrow, in 5 years, never?
- How will all this impact the system of work and our world?
- Do we need regulation — now, or later?
- Geopolitical implications (China, US, Europe)

Responsible Use & Critical Thinking

- Humans still matter
- Ethics, misinformation, responsibility, privacy, trust



Wrap-Up & Next Steps

Key Takeaways

- AI predicts patterns, doesn't "understand"
- Prompting is the valuable skill
- Powerful tool with real limitations
- Still early days of revolution

Your Homework

- Try AI tools (at home also)
- Practice prompt engineering
- Stay curious but critical

Questions?